



Generative Modelling of Maritime AIS Trajectories with Retrieval-Augmented Transformers

*A Land-Aware, Water-Strict Route Generator
from Start, End and Vessel Type*

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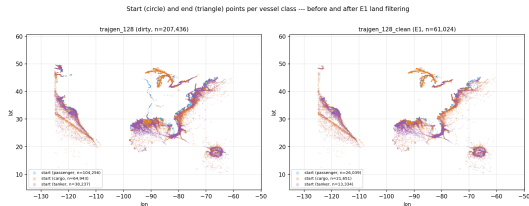
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Lab Research Project

Why learn maritime routes?

AIS broadcasts every large vessel's position - $\sim 10^8$ pings per year (NOAA, 2024)

- **Question:** given only (start, end, vessel type), can we sketch a plausible route?
- **Data:** 2024 US corpus $\approx 9.9 \times 10^7$ rows, $\sim 6 \times 10^5$ segments
- **Value:** traffic simulation, storm-risk routing, ETA, anomaly detection



Start (green) and end (red) distributions across the US coastal corpus

Problem: constrained full-route generation

Generate $T = 128$ waypoints minimising ADE under two constraints

■ Input $\rightarrow$ output:

$$q = (p_1, p_T, v) \rightarrow \hat{\tau} = (\hat{p}_1, \dots, \hat{p}_{128})$$

■ Objective: minimise

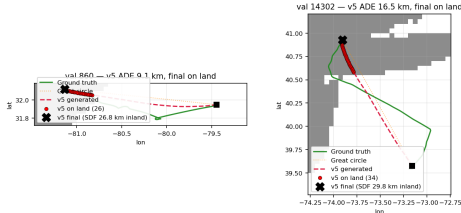
$$\text{ADE} = \frac{1}{T} \sum_t \text{haversine}(\hat{p}_t, p_t)$$

■ C1 - endpoint-exact: $\hat{p}_1 = p_1, \hat{p}_T = p_T$

■ C2 - water-valid: $\text{sdf}(\hat{p}_t) \leq \theta$ for all t

The three objectives pull against each other; baseline = great-circle geodesic.

v5 generates routes whose final waypoint terminates on land --- motivates v9/v10

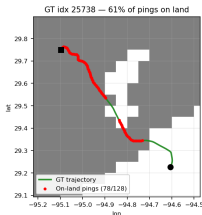
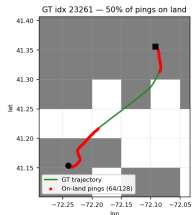


A model that ignores C2 runs aground on the coast

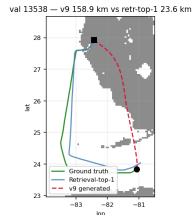
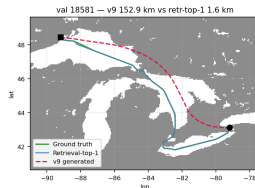
Four reasons it is hard

- **Irregular constraint surface:** land/water flips over a few km; narrow inlets need 0.005°
- **Multimodal routes:** many valid paths per query - ADE penalises the “wrong” valid mode
- **Two orders of magnitude:** 20 km hops ... 2000 km voyages - no single bias fits both
- **Noisy ground truth:** even cleaned GT crosses land 10.83% @10km before filtering

Dirty GT trajectories on inland water — motivates the E1 NPZ filter

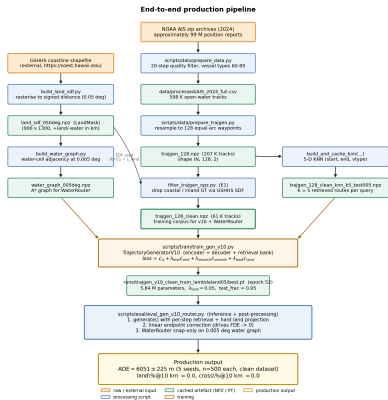


v9 mean-pool collapses on long routes (retrieval-top-1 outperforms it) — motivates v10 per-step retrieval



End-to-end production pipeline

Raw NOAA archives $\rightarrow$ *cleaned 128-waypoint corpus* $\rightarrow$ *trained model* $\rightarrow$ *post-processed route*

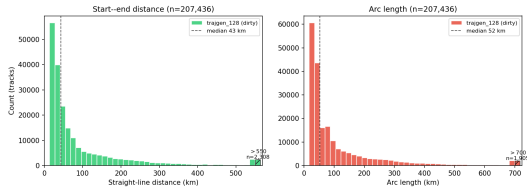
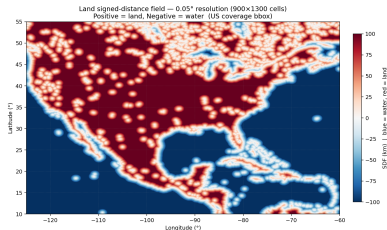


- **Orange:** raw / external input
- **Blue:** processing scripts
- **Green:** cached artefacts (built once)
- **Yellow:** training
- **Gold:** production output

Data quality: the single largest intervention

E1 land filter on the GSHHS shoreline → leak-free clean corpus

- **Problem:** unfiltered 128-pt GT crosses land 10.83% @10 km
- **Filter:** drop tracks with $>2\%$ land fraction or >10 km penetration $\Rightarrow$ 61k/207k kept
- **Impact:** larger than any single architecture change

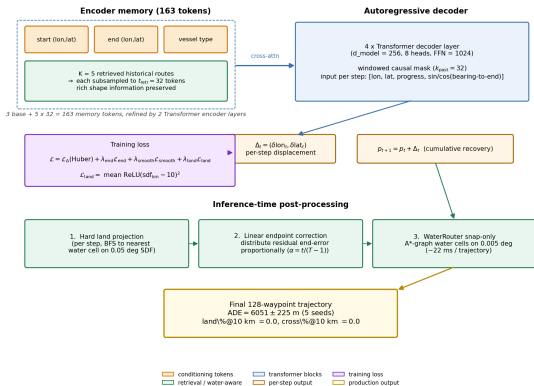


Corpus distributions after cleaning

Signed-distance field: + = land, - = water

Model architecture (v10)

Retrieval-augmented encoder-decoder - the production model

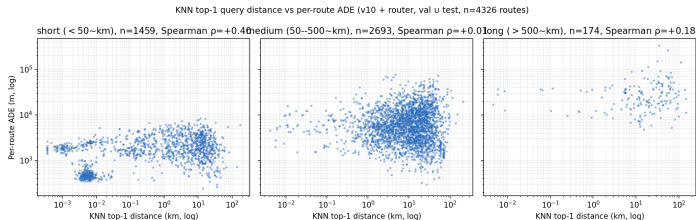


- **163 memory tokens:** 3 conditioning + 5×32 retrieval
- **Windowed-causal decoder:** one displacement/step, $k_{\text{past}} = 32$
- **Four-term loss:** Huber + endpoint + smoothness + land
- **Three inference passes:** project $\rightarrow$ endpoint fix $\rightarrow$ snap

Key idea 1 - per-step retrieval

The largest single architectural lever: >25% ADE reduction

- **How:** $K = 5$ historical neighbours $\times t_{\text{retr}} = 32$ steps as memory tokens
- **Why it beats v9:** decoder cross-attends *per step* - no mean-pool shape collapse
- **Evidence:** ADE correlates with top-1 neighbour distance ($\rho = +0.40$, short bucket)
- **Lesson:** try non-parametric memory before scaling the parametric model



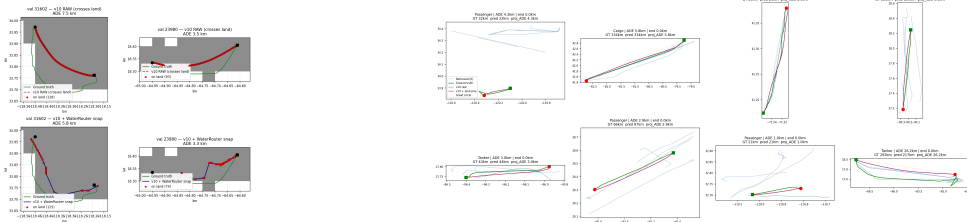
Per-route ADE vs nearest-neighbour distance - retrieval is doing real work

Key idea 2 - land-aware loss & WaterRouter snap

Differentiable SDF penalty in training; snap-to-water at inference $\rightarrow$ 0% crossings

- **Training:** $L_{\text{land}} = \text{meanReLU}(\text{sdf} - \theta)^2$ via bilinear grid_sample
- **Inference:** snap any violating waypoint to nearest *connected* water cell (0.005° graph)
- **Cost:** ~ 22 ms/route, no ADE penalty on the clean corpus

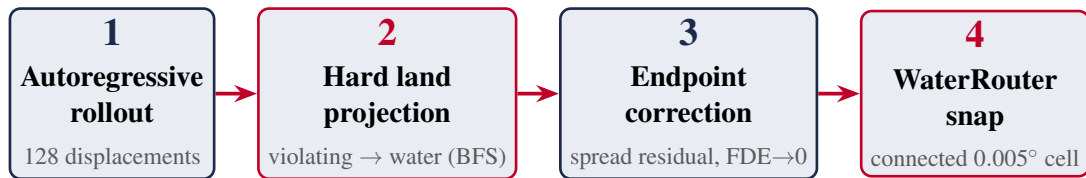
v10 raw (BT cuts across canal land masses); WaterRouter snaps the offending points back to water — evaluates the post-processor



Raw v10 output still clips a headland ...

... WaterRouter snaps it back onto water

Inference: three post-processing passes

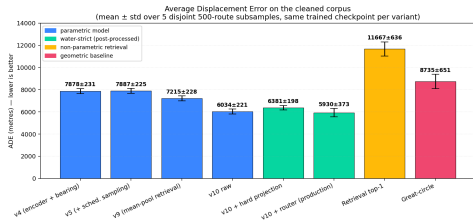


All four passes are post-hoc - no retraining - and together satisfy C1 (endpoint) and C2 (water-strict).

Headline results

v10 + router: -32% ADE vs great-circle, 0.0% land crossings

Method	ADE (m)	FDE (m)	cross% @10 km
v10 + router (prod.)	5930±372	608	0.04
v10 raw (same ckpt)	5913±384	0	1.88
Great-circle + snap	8484±597	581	0.20
Great-circle baseline	8735±651	0	8.80
Retrieval top-1 (0-train)	11667±636	11832	0.00



Setup: 5-seed mean ± std, disjoint 500-route subsamples

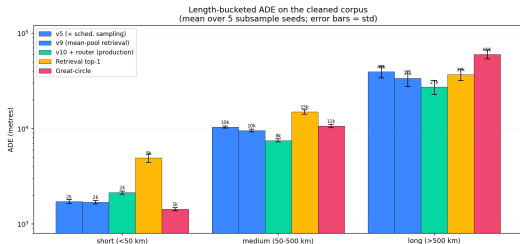
Best ADE: -32% vs great-circle

Water-strict: only learned method at 0% crossings

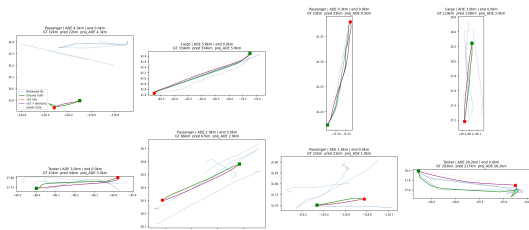
Free validity: router costs +17 m ADE vs raw

Where the model wins - and qualitative routes

Largest gains on medium/long voyages; great-circle still best on very short hops



ADE by route-length bucket

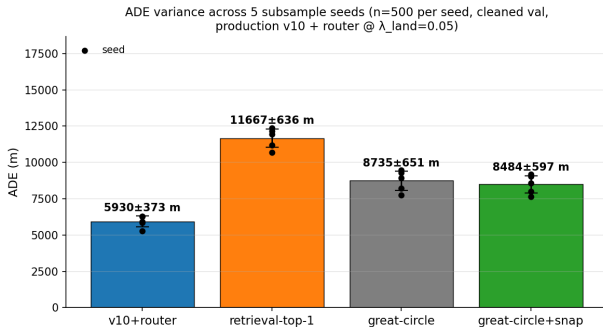


Generated routes (blue) vs ground truth

Long (>500 km): 27.4 km vs great-circle 60.1 km - learned shipping-lane shape; **short:** geodesic competitive.

Robustness: stable across seeds

The ranking and water-validity hold across 5 disjoint subsample seeds



ADE across 5 subsample seeds - v10+router vs baselines

- **Tight spread:** production ADE ± 225 – 372 m across seeds
- **Ranking never flips:** v10+router lowest in every seed
- **Always water-strict:** 0% crossings in all 5 seeds
- **Significant:** paired bootstrap CI vs great-circle excludes 0

Lessons, limitations & outlook

- **Retrieval = biggest architectural lever:** non-parametric memory beat 7 parametric ideas
- **Data quality = biggest single intervention:** water-strict ADE 8898 $\rightarrow$ 6051 m (same weights, dirty $\rightarrow$ clean eval); crossings $\rightarrow$ 0
- **Per-step loss $\neq$ rollout ADE:** only end-to-end evaluation should decide architecture

Limitations

- Generator for *known* waters - interpolates seen patterns
- Accuracy depends on retrieval-neighbour density
- No first-principles novel-geography reasoning

Outlook

- v13 - whole-sequence diffusion Transformer (EDM + CFG)
- Score-based land guidance during sampling
- Cross-region transfer & multimodal sampling



Thank you

Questions welcome

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